

CSAI321 – Computer Networks

Contest Exam Practice Sheet

120 MCQs + 20 Scenario-Based Questions, with Answers Attached

How to use this sheet: Every MCQ has its correct answer and a short explanation directly underneath it (in a green box) – cover it with your hand while attempting, then check. Every scenario question likewise has its full written solution directly underneath it (in a blue box). Covers all six lectures: Internet Fundamentals, Packet Switching & Delay, OSI/TCP-IP Layering, Devices & Topologies, HTTP/REST, and HTTPS/TLS. Good luck for the test!

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Part A: Multiple Choice Questions (120 Questions)

Lecture 1: How the Internet Works

1. The Internet is best described as:

- (A) A single powerful computer that hosts all websites
- (B) A network of networks connecting millions of devices
- (C) A private network owned by AWS
- (D) A single cable connecting all countries

Answer: B. The Internet is a network of networks – thousands of separate networks interconnected, with no single organization controlling all of it.

2. Which of these is part of the network edge?

- (A) Core router in an ISP backbone
- (B) Your laptop or smartphone
- (C) A submarine cable repeater
- (D) A BGP routing table

Answer: B. The network edge consists of end systems (devices that use the Internet) such as phones, laptops and servers.

3. The network core is best described as:

- (A) The collection of end-user devices
- (B) A single central data center
- (C) A mesh of routers that forward data from one place to another
- (D) The physical location of a cloud region

Answer: C. The network core is the mesh of interconnected routers whose job is to forward packets; it is not used directly by end users.

4. A LAN (Local Area Network) typically covers:

- (A) An entire country
- (B) A city
- (C) A small area like a home or office
- (D) The whole Internet

Answer: C. LAN = Local Area Network, used for small areas such as a home or office (e.g., your WiFi).

5. A MAN (Metropolitan Area Network) typically spans:

- (A) A single room
- (B) A city
- (C) A continent
- (D) 1-2 meters around a device

Answer: B. MAN = Metropolitan Area Network, spans a city-wide area, often used by municipalities or large organizations.

6. Which network type is the Internet itself best classified as?

- (A) PAN
- (B) LAN
- (C) MAN
- (D) WAN

Answer: D. The Internet spans the entire globe, so it is best classified as a WAN (Wide Area Network) – in fact, a network of networks.

7. Bluetooth headphones connecting to your phone is an example of a:

- (A) WAN
- (B) MAN
- (C) PAN
- (D) LAN

Answer: C. PAN = Personal Area Network, used for very short-range connections (1-2 meters), e.g., Bluetooth headphones.

8. Arrange these network types from smallest to largest geographical scope:

- (A) WAN, MAN, LAN, PAN
- (B) PAN, LAN, MAN, WAN
- (C) LAN, PAN, WAN, MAN
- (D) MAN, WAN, PAN, LAN

Answer: B. Correct scope ordering (smallest to largest): PAN -> LAN -> MAN -> WAN.

9. Approximately how many submarine cable systems connected continents as of 2025 (per lecture notes)?

- (A) 97
- (B) 250
- (C) 597+
- (D) 1,200

Answer: C. The notes state there were 597+ submarine cable systems in 2025 connecting continents.

10. Which of the following are commonly cited threats to submarine cables?

- (A) Software bugs and DNS spoofing
- (B) Fishing anchors and sharks
- (C) Power outages only
- (D) DDoS attacks

Answer: B. Fishing anchors/trawlers and sharks are the physical threats highlighted for submarine cables.

11. An AWS 'Region' is best described as:

- (A) A single physical server
- (B) A geographic area containing multiple data centers
- (C) A CDN cache point close to the user
- (D) A private virtual network inside AWS

Answer: B. A Region is a geographic area (e.g., Mumbai Region) that contains multiple data centers.

12. An 'Availability Zone (AZ)' is best described as:

- (A) The entire AWS global network
- (B) One isolated fault domain (one or more data centers) within a Region
- (C) A CDN edge cache
- (D) A firewall rule set

Answer: B. An AZ is one or more data centers within a Region, acting as an isolated fault domain – if one fails, others can keep running.

13. Edge Locations in AWS primarily help reduce:

- (A) Encryption overhead
- (B) Latency, by caching content closer to the user
- (C) The number of Availability Zones needed
- (D) The cost of storage

Answer: B. Edge Locations are CDN endpoints that cache content near users, reducing propagation delay/latency.

14. A VPC (Virtual Private Cloud) is:

- (A) A physical router in a data center
- (B) Your own private, software-defined network inside AWS
- (C) A public DNS server
- (D) A type of submarine cable

Answer: B. A VPC lets you create your own private, software-defined network inside AWS, controlling IP ranges, subnets, and routing.

15. Latency is best defined as:

- (A) The maximum bandwidth of a link
- (B) The time taken for a request's response to arrive back
- (C) The number of hops in a route
- (D) The size of a packet in bits

Answer: B. Latency = how much time it takes for a request to get a response; more hops generally means more latency.

16. In the 'Instagram tap' journey described in the notes, which comes right after the request leaves your router?

- (A) AWS/Meta servers directly
- (B) The ISP
- (C) Another user's phone
- (D) A submarine cable repeater only

Answer: B. The described flow is: Thumb tap -> App opens -> WiFi/4G -> Router -> ISP -> Internet Core -> AWS/Meta servers -> data returns.

17. Which statement about who controls the Internet is TRUE?

- (A) A single global organization controls all of it
- (B) No single organization controls it; it is a federation of many networks
- (C) Only the UN controls it
- (D) Only AWS, Azure and GCP control it

Answer: B. The Internet is a federation of lakhs of networks – ISPs, companies and governments – with no single controlling organization.

18. Which of these is NOT one of the three major cloud providers mentioned in the notes?

- (A) Amazon Web Services (AWS)
- (B) Microsoft Azure
- (C) Google Cloud Platform (GCP)
- (D) Oracle Cloud

Answer: D. The notes name AWS, Azure and GCP as the three major cloud providers discussed.

Lecture 2: Moving Data Through the Core

19. In circuit switching, network resources for a call/session are:

- (A) Shared dynamically among all users
- (B) Reserved exclusively for the entire duration of the session
- (C) Never reserved at all
- (D) Reserved only for the first second

Answer: B. Circuit switching reserves a dedicated path for the whole session, even if it's idle (e.g., during silence on a call).

20. The main disadvantage of circuit switching for bursty traffic is:

- (A) It is too fast
- (B) Idle reserved capacity is wasted
- (C) It cannot guarantee bandwidth
- (D) It requires no setup time

Answer: B. Because capacity is reserved for the whole call, any pause (like silence) wastes that reserved, unused capacity.

21. In packet switching, data is:

- (A) Sent as one continuous stream over a reserved path
- (B) Broken into independent packets that can take different routes
- (C) Never broken into smaller units
- (D) Only sent after the whole path is reserved

Answer: B. Packet switching splits data into independent packets, each of which may take a different route to the same destination.

22. A packet consists of:

- (A) Only a header
- (B) Only a payload

- (C) A header (addressing info) and a payload (actual data)
- (D) A payload and a certificate

Answer: C. Packet anatomy = Header (source + destination address) + Payload (the actual data).

23. The store-and-forward rule at a router means:

- (A) A router forwards bits as soon as the first bit arrives
- (B) A router must fully receive the entire packet before forwarding any of it
- (C) A router stores packets permanently
- (D) A router only forwards packets larger than 1500 bytes

Answer: B. Store-and-forward: not a single bit is forwarded until the whole packet has arrived at the router.

24. Because of store-and-forward and hop-by-hop routing, a router needs to know:

- (A) The complete end-to-end path to the destination
- (B) Only the next hop
- (C) The IP addresses of every device on the Internet
- (D) Nothing about addressing

Answer: B. Each router only needs to know the next hop; it does not need to know the entire path to the destination.

25. Statistical multiplexing gives packet-switched networks which two advantages?

- (A) Guaranteed bandwidth and zero delay
- (B) Efficiency and resilience
- (C) Lower cost hardware and higher latency
- (D) No packet loss and no jitter

Answer: B. Because links aren't reserved, they can be efficiently shared (efficiency) and traffic can reroute around failures (resilience).

26. Which delay type involves the router reading the packet header and checking for bit errors?

- (A) Queuing delay
- (B) Processing delay
- (C) Transmission delay
- (D) Propagation delay

Answer: B. Processing delay is the (tiny, ~microsecond) time a router takes to examine the header and check for errors.

27. Queuing delay in this course is:

- (A) Calculated using L/R
- (B) Calculated using d/s
- (C) Variable and out of scope for exam calculations
- (D) Always exactly zero

Answer: C. Queuing delay is variable (depends on congestion) and its mathematics is explicitly marked out of exam scope – only conceptual understanding is needed.

28. **Transmission delay is calculated using the formula:**

- (A) d / s
- (B) L / R
- (C) $n(n-1)/2$
- (D) R / L

Answer: B. Transmission delay = L/R , where L is packet size in bits and R is the link rate in bits/sec.

29. **Propagation delay is calculated using the formula:**

- (A) L / R
- (B) d / s
- (C) $\min(R1, R2, R3)$
- (D) $n / 2$

Answer: B. Propagation delay = d/s , where d is distance and s is the propagation speed of the signal ($\sim 3 \times 10^8$ m/s in fiber/air).

30. **A packet of 8000 bits is sent over a 1 Mbps link. What is the transmission delay?**

- (A) 0.8 ms
- (B) 8 ms
- (C) 80 ms
- (D) 8 seconds

Answer: B. Transmission delay = $L/R = 8000 / 1,000,000 = 0.008$ s = 8 ms.

31. **A packet of 4000 bits is sent over a 2 Mbps link. What is the transmission delay?**

- (A) 0.5 ms
- (B) 2 ms
- (C) 4 ms
- (D) 8 ms

Answer: B. Transmission delay = $L/R = 4000 / 2,000,000 = 0.002$ s = 2 ms.

32. **If a signal travels 2×10^6 meters through a medium at 2×10^8 m/s, the propagation delay is:**

- (A) 0.001 s (1 ms)
- (B) 0.01 s (10 ms)
- (C) 0.1 s (100 ms)
- (D) 1 s

Answer: B. Propagation delay = $d/s = (2 \times 10^6) / (2 \times 10^8) = 0.01$ s = 10 ms.

33. **If you double the link rate R while keeping packet size L constant, the transmission delay will:**

- (A) Double
- (B) Stay the same
- (C) Halve
- (D) Become zero

Answer: C. Since transmission delay = L/R , doubling R (with L fixed) halves the transmission delay – an inverse relationship.

34. **A satellite link advertises 500 Mbps bandwidth but users still notice a long delay. This is best explained by:**

- (A) Low bandwidth
- (B) High propagation delay due to the large physical distance to the satellite and back
- (C) Packet loss only
- (D) Encryption overhead

Answer: B. High bandwidth does not mean low latency – propagation delay depends on distance/speed, not on bandwidth; a geostationary satellite round trip is large regardless of link speed.

35. **End-to-end throughput across a multi-hop path is approximately:**

- (A) The average of all link rates
- (B) The sum of all link rates
- (C) The minimum (bottleneck) link rate along the path
- (D) The maximum link rate along the path

Answer: C. Throughput = $\min(R_1, R_2, R_3, \dots)$ – the slowest (bottleneck) link on the path determines the end-to-end rate.

36. **A path has links of 100 Mbps, 10 Mbps, and 50 Mbps in series. What is the approximate end-to-end throughput?**

- (A) 100 Mbps
- (B) 53 Mbps (average)
- (C) 10 Mbps (bottleneck)
- (D) 160 Mbps (sum)

Answer: C. Throughput is limited by the slowest link (bottleneck) in the path, which is 10 Mbps here.

37. **Packet loss typically occurs when:**

- (A) A router's buffer/queue is full and new packets are dropped
- (B) The propagation delay is too low
- (C) The packet header is too small
- (D) Two packets have the same source IP

Answer: A. When router buffers fill up (queue full), incoming new packets have nowhere to go and are dropped – this is packet loss.

38. **Which protocol/layer is responsible for detecting packet loss and retransmitting lost data (mentioned as a later topic)?**

- (A) IP at the Network layer

- (B) TCP at the Transport layer
- (C) HTTP at the Application layer
- (D) MAC at the Data Link layer

Answer: B. TCP (Transport layer) later handles detecting loss, retransmission, and pacing to make the unreliable core feel reliable.

39. Which of the four delay components does NOT need to be manually calculated for this exam?

- (A) Transmission delay
- (B) Propagation delay
- (C) Queuing delay
- (D) Both A and B

Answer: C. Only Transmission (L/R) and Propagation (d/s) delays are calculated in this course; Queuing delay math is out of scope.

40. The main reason networks use a layered design is:

- (A) It makes the network slower but more secure
- (B) A network is too complex to design/debug as one single unit, so each layer handles one job
- (C) It reduces the number of protocols needed to just one
- (D) Layers must always know what every other layer is doing

Answer: B. Layering breaks an overwhelming problem into manageable pieces – each layer solves one problem and talks only to layers directly above/below it.

Lecture 3: Protocol Layers (OSI & TCP/IP)

41. OSI stands for:

- (A) Open System Interconnection
- (B) Optimal Signal Interface
- (C) Online Server Infrastructure
- (D) Open Source Internet

Answer: A. OSI = Open Systems Interconnection, a 7-layer conceptual model developed by ISO.

42. How many layers does the OSI model have?

- (A) 4
- (B) 5
- (C) 6
- (D) 7

Answer: D. The OSI model has 7 layers: Physical, Data Link, Network, Transport, Session, Presentation, Application.

43. How many layers does the TCP/IP model have?

- (A) 4
- (B) 5
- (C) 7

(D) 3

Answer: A. The TCP/IP model condenses networking into 4 layers: Link/Network Access, Internet/Network, Transport, and Application.

44. **Which OSI layer is responsible for routing and logical (IP) addressing across different networks?**

- (A) Layer 2 - Data Link
- (B) Layer 3 - Network
- (C) Layer 4 - Transport
- (D) Layer 6 - Presentation

Answer: B. Layer 3 (Network) handles logical/IP addressing and routing packets across different networks – the 'whole journey'.

45. **Which OSI layer uses MAC addresses and operates within a single LAN (one hop)?**

- (A) Layer 1 - Physical
- (B) Layer 2 - Data Link
- (C) Layer 3 - Network
- (D) Layer 5 - Session

Answer: B. Layer 2 (Data Link) uses MAC addresses for delivery within one LAN and does error checking – scope is one hop, not the whole journey.

46. **Encryption/decryption and data compression are primarily handled at which OSI layer?**

- (A) Layer 4 - Transport
- (B) Layer 5 - Session
- (C) Layer 6 - Presentation
- (D) Layer 7 - Application

Answer: C. Layer 6 (Presentation) handles encryption/decryption, compression, and character/format encoding conversions.

47. **Which OSI layer is responsible for reliable end-to-end delivery (e.g., TCP/UDP)?**

- (A) Layer 3 - Network
- (B) Layer 4 - Transport
- (C) Layer 2 - Data Link
- (D) Layer 1 - Physical

Answer: B. Layer 4 (Transport) is where TCP/UDP operate, providing end-to-end delivery and reliability.

48. **The Protocol Data Unit (PDU) at the Transport layer is called a:**

- (A) Frame
- (B) Packet
- (C) Segment
- (D) Bit

Answer: C. Transport layer PDU = Segment (TCP/UDP 'segment' the data).

49. **The Protocol Data Unit (PDU) at the Network layer is called a:**

- (A) Segment
- (B) Packet
- (C) Frame
- (D) Message

Answer: B. Network layer PDU = Packet.

50. **The Protocol Data Unit (PDU) at the Data Link layer is called a:**

- (A) Packet
- (B) Frame
- (C) Bits
- (D) Segment

Answer: B. Data Link layer PDU = Frame.

51. **The Protocol Data Unit (PDU) at the Physical layer is called:**

- (A) Bits
- (B) Frame
- (C) Packet
- (D) Message

Answer: A. Physical layer PDU = Bits (the actual 0s and 1s transmitted as signals).

52. **Encapsulation refers to the process where:**

- (A) Each layer strips headers as data moves down the sender's stack
- (B) Each layer adds its own header as data moves down the sender's stack
- (C) Only the top layer adds a header
- (D) Headers are added at the receiver, not the sender

Answer: B. On the sender side (moving down the stack), each layer wraps/encapsulates the data from the layer above by adding its own header.

53. **Decapsulation refers to the process where:**

- (A) The receiver adds headers going up the stack
- (B) The receiver strips each layer's header as data moves up the stack
- (C) The sender removes headers before sending
- (D) Headers are never removed

Answer: B. On the receiver side (moving up the stack), each layer strips its corresponding header (decapsulation) before passing data further up.

54. **Why does a switch not need to understand IP addresses?**

- (A) Because switches operate at Layer 2 using MAC addresses, not Layer 3
- (B) Because IP addresses do not exist on LANs
- (C) Because switches are Layer 4 devices

(D) Because switches only use physical signals with no addressing

Answer: A. Switches operate at Layer 2 (Data Link) and forward frames using MAC addresses; they don't need Layer 3 IP addressing.

55. The famous OSI mnemonic 'All People Seem To Need Data Processing' represents the layers from:

- (A) Layer 1 to Layer 7 (bottom to top)
- (B) Layer 7 to Layer 1 (top to bottom)
- (C) Only the top 3 layers
- (D) Only the bottom 3 layers

Answer: B. This mnemonic goes top-down: Application, Presentation, Session, Transport, Network, Data Link, Physical.

56. In the TCP/IP model, the 'Application' layer combines which OSI layers?

- (A) Layers 1, 2, 3
- (B) Layers 5, 6, 7
- (C) Layers 3, 4
- (D) Only Layer 7

Answer: B. TCP/IP's Application layer maps to OSI's Session, Presentation and Application layers (5, 6, 7).

57. Which best captures the Layer 2 vs Layer 3 distinction (per the notes)?

- (A) Layer 2 = across the whole world; Layer 3 = one building only
- (B) Layer 2 = delivery within one LAN (MAC address); Layer 3 = delivery across networks/city-to-city (IP address)
- (C) Layer 2 and Layer 3 are identical
- (D) Layer 2 handles encryption, Layer 3 handles compression

Answer: B. Layer 2 = 'gali ke andar delivery' (within one LAN, MAC address); Layer 3 = 'city to city delivery' (across networks, IP address).

58. In the letter/postal analogy for layering, why doesn't the postman need to know the letter's content?

- (A) Because postmen are not allowed to read anything ever
- (B) Because each layer only needs the info relevant to its own job (the address), not the content of layers above it
- (C) Because letters never have content
- (D) Because the postman is actually the sender

Answer: B. Just like each layer cares only about its own header info (e.g., an address), the postman only needs the address – not the letter's content – to deliver it.

59. Which of the following is a direct benefit of layering (per the notes)?

- (A) You can replace one layer's technology (e.g., WiFi to Ethernet) without rewriting the layers above it
- (B) It eliminates the need for any protocols

- (C) It guarantees zero packet loss
- (D) It removes the need for addressing

Answer: A. Layering allows independent replacement of a layer's implementation (e.g., swapping WiFi for Ethernet) without touching the other layers.

60. A hub operates at which OSI layer and how does it handle incoming data?

- (A) Layer 2; forwards only to the correct MAC address
- (B) Layer 1; broadcasts incoming data to every connected port
- (C) Layer 3; routes based on IP address
- (D) Layer 4; inspects port numbers

Answer: B. A hub is a 'dumb' Layer 1 device – it broadcasts whatever it receives on one port to all other ports, with no addressing intelligence.

Lecture 4: Devices, Topologies & Cloud Networking

61. A switch operates at which OSI layer, and what does it use to make forwarding decisions?

- (A) Layer 1; electrical signals
- (B) Layer 2; MAC addresses
- (C) Layer 3; IP addresses
- (D) Layer 4; port numbers

Answer: B. A switch operates at Layer 2 (Data Link) and forwards frames intelligently using learned MAC address-to-port mappings.

62. A router operates at which OSI layer, and what does it use for routing decisions?

- (A) Layer 1; raw bits
- (B) Layer 2; MAC addresses
- (C) Layer 3; IP addresses
- (D) Layer 7; application data

Answer: C. A router operates at Layer 3 (Network) and routes packets between different networks using IP addresses.

63. A basic packet-filtering firewall typically inspects which layers?

- (A) Only Layer 1
- (B) Layer 2 only (MAC addresses)
- (C) Layer 3 and Layer 4 (IP addresses and port numbers)
- (D) Layer 7 only (application content)

Answer: C. A basic firewall checks source/destination IP (Layer 3) and source/destination port numbers (Layer 4) to decide whether to allow or block traffic.

64. A load balancer typically operates at which layer(s) and does what?

- (A) Layer 1; repeats signals
- (B) Layer 4/Layer 7; distributes incoming requests across multiple servers
- (C) Layer 2; learns MAC addresses
- (D) Layer 3; assigns IP addresses

Answer: B. A load balancer (L4/L7) decides which server should receive a given request, distributing traffic to avoid overloading a single server.

65. The number of cables needed to directly connect every pair among n devices (full mesh, no central device) is:

- (A) n
- (B) n^2
- (C) $n(n-1)/2$
- (D) $2n$

Answer: C. The direct-connection formula is $n(n-1)/2$, which grows quadratically as devices increase – this is why switches/hubs are used instead.

66. How many cables are needed to directly connect 10 devices to each other (full mesh)?

- (A) 10
- (B) 45
- (C) 90
- (D) 100

Answer: B. Using $n(n-1)/2 = 10 \times 9 / 2 = 45$ cables.

67. The main weakness of a Star topology is:

- (A) It requires too many cables ($n(n-1)/2$)
- (B) The central hub/switch is a single point of failure
- (C) It cannot be used for LANs
- (D) It has no central point at all

Answer: B. In a Star topology, if the central device fails, every connected device loses connectivity – a single point of failure (SPOF).

68. Which topology offers the highest fault tolerance because of multiple redundant paths?

- (A) Star
- (B) Ring
- (C) Mesh
- (D) Bus

Answer: C. Mesh topology connects every device directly to every other device, giving very high fault tolerance via redundant paths (at the cost of many cables).

69. The main weakness of Ring topology is:

- (A) Too expensive in cabling ($n(n-1)/2$)
- (B) A single break in the ring can bring down the entire network
- (C) It requires a central switch
- (D) It cannot be scaled at all

Answer: B. In a Ring topology, all devices are connected in a loop – if one connection breaks, the whole ring can fail.

70. Which topology is described as 'best of all' because it combines multiple topology types and is commonly used in data centers?

- (A) Star
- (B) Mesh
- (C) Ring
- (D) Hybrid

Answer: D. Hybrid topology combines multiple topology types, offering flexibility; real-world large networks (like data centers) typically use hybrid designs.

71. According to the environment-to-topology mapping in the notes, which topology suits a small office best?

- (A) Mesh
- (B) Star
- (C) Ring
- (D) Hybrid

Answer: B. Star topology is recommended for small offices – easy setup and easy debugging.

72. According to the environment-to-topology mapping in the notes, which topology suits a smart home best?

- (A) Star
- (B) Hybrid
- (C) Mesh
- (D) Ring

Answer: C. Mesh topology suits smart homes – devices like smart bulbs and cameras benefit from multiple redundant direct paths (e.g., wireless mesh).

73. An AWS Region containing 30+ deployments globally is best described as:

- (A) A single server rack
- (B) An isolated geographic area with multiple data centers
- (C) A CDN edge cache
- (D) A firewall configuration

Answer: B. A Region is an isolated geographic area (e.g., Mumbai) containing multiple data centers, and there are 30+ globally.

74. If a fire occurs in one Availability Zone but the application keeps running using another AZ, this demonstrates:

- (A) Statistical multiplexing
- (B) Multi-AZ deployment providing fault tolerance
- (C) Head-of-line blocking
- (D) Store-and-forward routing

Answer: B. Deploying across multiple AZs means a localized failure (fire, flood, power outage) in one AZ doesn't take down the whole application – this is the point of AZs as isolated fault domains.

75. Edge Locations reduce latency mainly because:

- (A) They increase the link's bandwidth (R)
- (B) They reduce the physical distance (d) data has to travel, lowering propagation delay
- (C) They encrypt data faster
- (D) They eliminate transmission delay entirely

Answer: B. Since propagation delay = d/s , caching content physically closer to the user (smaller d) directly reduces propagation delay.

76. A VPC lets an AWS customer control:

- (A) Only the physical cabling in the data center
- (B) IP address ranges, subnets, and routing within their private network
- (C) The Earth's rotation of satellites
- (D) Only DNS records

Answer: B. A VPC (Virtual Private Cloud) is a private, software-defined network where the customer controls IP ranges, subnets, and routing.

77. If a server (not a network link) is the bottleneck causing slow responses, the best fix (per the notes) is to:

- (A) Increase transmission delay
- (B) Add more servers and use a load balancer to distribute traffic
- (C) Switch from packet switching to circuit switching
- (D) Reduce the propagation delay

Answer: B. Rather than making one server faster, the fix is to add more servers and use a load balancer to distribute traffic across them.

78. Compared to a hub, a switch is often described as:

- (A) A hub with memory (it learns MAC-to-port mappings)
- (B) A hub with less intelligence
- (C) Identical to a hub in every way
- (D) A device that operates only at Layer 1

Answer: A. A switch is like 'a hub with memory' – it learns which MAC address is on which port, so it forwards frames only where needed instead of broadcasting.

79. A wireless router/access point typically operates across which layers (per the notes)?

- (A) Only Layer 1
- (B) Layers 1 to 3
- (C) Only Layer 7
- (D) Layers 5 to 7

Answer: B. A wireless router/AP provides wireless access and routing functionality, spanning roughly Layers 1-3.

80. In a Client-Server architecture:

- (A) All devices act as equal peers with no dedicated server
- (B) Powerful always-on servers store data and serve multiple clients
- (C) There is no central point of control
- (D) Clients directly communicate with each other, bypassing servers

Answer: B. Client-Server is centralized: always-on servers store/process data and multiple clients depend on them (e.g., YouTube, Gmail).

Lecture 5: HTTP, P2P & REST

81. In a Peer-to-Peer (P2P) architecture:

- (A) There is one always-on central server
- (B) Every peer acts as both a client and a server, and there's no central server
- (C) Only the server can upload data
- (D) It cannot scale at all

Answer: B. P2P is decentralized: every peer can act as both client and server, e.g., BitTorrent – it scales well but is harder to manage.

82. HTTP operates at which OSI/TCP-IP layer?

- (A) Physical layer
- (B) Transport layer
- (C) Application layer
- (D) Data Link layer

Answer: C. HTTP (HyperText Transfer Protocol) is an Application layer protocol – the 'language' of the web.

83. HTTP is described as 'stateless', meaning:

- (A) The server remembers every previous request from a client automatically
- (B) The server does not automatically remember previous requests
- (C) HTTP requests never have a body
- (D) HTTP cannot be encrypted

Answer: B. Stateless means each HTTP request/response is independent; the server does not inherently remember earlier requests.

84. Since HTTP is stateless, how do features like a shopping cart persist across pages?

- (A) They don't – carts reset on every click
- (B) Through cookies and session tokens stored by the browser and sent with each request
- (C) Through circuit switching
- (D) Through the OSI Presentation layer only

Answer: B. Cookies/session tokens let the browser store an identifier that is sent with every request, letting the server 'recognize' the client despite statelessness.

85. An HTTP request line typically contains:

- (A) Status code, status message, HTTP version
- (B) Method, URL, HTTP version
- (C) Only the URL
- (D) Content-Type and Content-Length only

Answer: B. The request line has three parts: [Method] [URL] [HTTP Version], e.g., GET /index.html HTTP/1.1.

86. An HTTP response status line typically contains:

- (A) Method, URL, HTTP version
- (B) HTTP version, status code, status message
- (C) Only the response body
- (D) Host header only

Answer: B. The status line has three parts: HTTP version, a 3-digit status code, and a short status message (e.g., HTTP/1.1 200 OK).

87. Which HTTP method is considered 'safe' (no side effects, does not change server state)?

- (A) POST
- (B) PUT
- (C) GET
- (D) DELETE

Answer: C. GET only reads/retrieves data and has no side effects, making it 'safe' – unlike POST/PUT/DELETE which modify state.

88. Which HTTP method is used to CREATE a new resource (CRUD mapping)?

- (A) GET
- (B) POST
- (C) PATCH
- (D) DELETE

Answer: B. POST maps to Create in CRUD – used to submit/create new data on the server.

89. Which HTTP method fully replaces an existing resource?

- (A) GET
- (B) POST
- (C) PUT
- (D) PATCH

Answer: C. PUT is used to Update by replacing the existing resource entirely.

90. Which HTTP method updates only specific fields of a resource, rather than the whole thing?

- (A) PUT
- (B) PATCH
- (C) GET
- (D) DELETE

Answer: B. PATCH performs a partial update – only the specified fields are changed, unlike PUT which replaces everything.

91. HTTP status codes in the 2xx range indicate:

- (A) Client error
- (B) Server error
- (C) Success

(D) Redirection

Answer: C. 2xx = Success, e.g., 200 OK (request successful) or 201 Created.

92. **HTTP status code 404 means:**

- (A) Server error
- (B) Resource not found
- (C) Redirect
- (D) Unauthorized but resource exists

Answer: B. 404 Not Found – the requested resource could not be found on the server.

93. **HTTP status code 500 means:**

- (A) Bad Request from the client
- (B) Not Found
- (C) Internal Server Error – the client was fine, but the server had a problem
- (D) Successful creation

Answer: C. 500 Internal Server Error indicates the client's request was fine, but something went wrong on the server side.

94. **What was the key limitation of HTTP/1.0 (1996)?**

- (A) It supported multiplexing
- (B) Every file required a brand-new TCP connection – no persistent connections
- (C) It used QUIC
- (D) It was fully binary

Answer: B. HTTP/1.0 opened a new connection per file/request, requiring the TCP setup to repeat every time – very slow and inefficient.

95. **HTTP/1.1's Keep-Alive feature allows:**

- (A) Multiple requests over a single persistent TCP connection
- (B) Requests to skip the TCP layer entirely
- (C) Automatic encryption of all traffic
- (D) Elimination of all delay types

Answer: A. HTTP/1.1 introduced persistent (Keep-Alive) connections, allowing multiple requests/responses over one connection instead of opening a new one each time.

96. **Head-of-Line (HoL) blocking in HTTP/1.1 means:**

- (A) Requests can be sent in any order with no consequences
- (B) Responses must arrive in FIFO order, so one slow response blocks everything behind it
- (C) Only the first request is ever processed
- (D) The server ignores all but the first client

Answer: B. Because responses must return in the same (FIFO) order as requests, one slow response delays all the responses queued behind it.

97. **Which HTTP methods are safe to use with HTTP/1.1 pipelining?**

- (A) POST and PUT only
- (B) GET, HEAD, and OPTIONS
- (C) DELETE and PATCH
- (D) All methods equally

Answer: B. Only safe methods (GET, HEAD, OPTIONS) can be pipelined; POST/PUT are excluded since they change server state (side effects).

98. **HTTP/2's key innovation of sending multiple streams over a single TCP connection simultaneously is called:**

- (A) Server Push
- (B) Multiplexing
- (C) Statelessness
- (D) Statistical multiplexing

Answer: B. HTTP/2 uses multiplexing – multiple streams share one TCP connection simultaneously, fixing application-level HoL blocking.

99. **Despite HTTP/2's multiplexing, what problem still remained?**

- (A) Application-level HoL blocking
- (B) TCP-level HoL blocking (one lost TCP segment stalls all streams)
- (C) No persistent connections
- (D) It could not use binary framing

Answer: B. HTTP/2 fixed HoL blocking at the application level, but because it still runs over a single TCP connection, one lost TCP segment can stall ALL streams (TCP-level HoL blocking).

100. **HTTP/3 solves the remaining HoL blocking problem by:**

- (A) Going back to HTTP/1.0-style connections
- (B) Using QUIC over UDP, giving truly independent streams
- (C) Removing multiplexing entirely
- (D) Using more TCP connections in parallel

Answer: B. HTTP/3 replaces TCP with QUIC (UDP-based); each stream is truly independent, so loss in one stream doesn't stall the others.

101. **A REST API primarily maps HTTP methods to which set of operations?**

- (A) Encrypt, Decrypt, Sign, Verify
- (B) CRUD: Create, Read, Update, Delete
- (C) Push, Pull, Merge, Commit
- (D) Connect, Bind, Listen, Accept

Answer: B. REST = HTTP methods + meaningful URLs = clean API design, mapping to CRUD (POST=Create, GET=Read, PUT/PATCH=Update, DELETE=Delete).

102. **The browser padlock icon guarantees:**

- (A) The website is completely safe and free of malware

- (B) The connection is encrypted and the server's identity has been verified – not that the content itself is safe
- (C) The website has no ads
- (D) The server is located in your own country

Answer: B. The padlock means the connection is encrypted and identity is CA-verified – it does NOT guarantee the site is free of malware or otherwise 'safe'.

Lecture 6: HTTPS & TLS

103. HTTP typically uses which port, and HTTPS uses which port?

- (A) HTTP: 443, HTTPS: 80
- (B) HTTP: 80, HTTPS: 443
- (C) HTTP: 21, HTTPS: 22
- (D) HTTP: 8080, HTTPS: 8443

Answer: B. HTTP uses port 80 (plain text); HTTPS uses port 443 (encrypted).

104. HTTPS can be summarized by the formula:

- (A) HTTP + FTP
- (B) HTTP + TLS
- (C) HTTP + SSH
- (D) HTTP + IP

Answer: B. HTTPS = HTTP + TLS (Transport Layer Security), providing encryption and authentication on top of HTTP.

105. Which statement about SSL and TLS is correct?

- (A) SSL is the modern replacement for TLS
- (B) TLS is the modern protocol that replaced the older SSL, though the name 'HTTPS' remains
- (C) SSL and TLS are unrelated protocols
- (D) TLS only works over UDP

Answer: B. SSL (Secure Socket Layer) was used earlier; TLS (Transport Layer Security) is now used, though the term HTTPS persists.

106. Confidentiality, as a cryptography goal, means:

- (A) Verifying who sent the message
- (B) Ensuring data isn't tampered with in transit
- (C) Ensuring only authorized parties can read the data (no eavesdropping)
- (D) Making data available 24/7

Answer: C. Confidentiality = no eavesdropping – only authorized parties can read the information, typically achieved via encryption.

107. Integrity, as a cryptography goal, means:

- (A) No eavesdropping
- (B) No tampering – data isn't modified in transit
- (C) Only authorized parties can connect
- (D) Fast data transfer

Answer: B. Integrity = no tampering – ensuring the data received is exactly what was sent, achieved via hashing/MACs.

108. **Authentication, as a cryptography goal, means:**

- (A) Encrypting the payload only
- (B) Compressing the data
- (C) Verifying the identity of the sender/receiver
- (D) Guaranteeing bandwidth

Answer: C. Authentication answers 'who are you exactly?' – verifying identities, typically achieved via certificates.

109. **In symmetric encryption:**

- (A) Two different keys (public/private) are used
- (B) The same key is used for both encryption and decryption
- (C) No key is required at all
- (D) Only the receiver has any key

Answer: B. Symmetric encryption uses one shared secret key for both encrypting and decrypting – fast, but both parties must have the same key.

110. **The main problem with symmetric encryption alone (for strangers communicating) is:**

- (A) It is too slow for any use
- (B) Securely sharing the secret key with a stranger without interception is hard
- (C) It cannot be used for real data at all
- (D) It requires a Certificate Authority

Answer: B. If two strangers have never met, safely exchanging the same secret key without an eavesdropper intercepting it is the key-distribution problem.

111. **In asymmetric encryption, data encrypted with the recipient's public key can only be decrypted with:**

- (A) Any public key
- (B) The recipient's matching private key
- (C) The sender's private key
- (D) A shared symmetric key

Answer: B. Asymmetric encryption uses a public key (shareable, used to encrypt) and a private key (secret, used to decrypt) – only the private key owner can decrypt.

112. **A key drawback of asymmetric encryption is that it is:**

- (A) Impossible to use for authentication
- (B) Slower than symmetric encryption, so it isn't used for bulk/real data
- (C) Never secure
- (D) Identical in speed to symmetric encryption

Answer: B. Asymmetric encryption is comparatively slow, so it's used for secure key exchange/handshakes, not for encrypting large amounts of real data.

113. **TLS's key insight of combining both encryption types works as:**

- (A) Symmetric for the handshake, asymmetric for the data
- (B) Asymmetric to safely agree on a shared secret; symmetric for fast actual data encryption
- (C) Only symmetric encryption is used throughout
- (D) Only asymmetric encryption is used throughout

Answer: B. TLS uses asymmetric crypto during the handshake to safely establish a shared session key, then switches to fast symmetric encryption for the actual data.

114. **A digital certificate primarily contains:**

- (A) The client's browsing history
- (B) The server's public key plus a digital signature from a Certificate Authority
- (C) The user's password in plaintext
- (D) A list of all websites the CA has ever signed

Answer: B. A certificate is the server's identity proof – it bundles the server's public key with the CA's digital signature.

115. **A Certificate Authority (CA) is best described as:**

- (A) A malicious third party
- (B) A trusted third party that verifies and signs certificates
- (C) The end-user's own device
- (D) A type of firewall

Answer: B. CAs (e.g., DigiCert, Let's Encrypt) are trusted third parties that verify server identities and sign their certificates.

116. **The 'chain of trust' in HTTPS ultimately relies on:**

- (A) Every user manually verifying each certificate by hand
- (B) Root CAs that are pre-installed and already trusted by the browser/OS
- (C) No trust at all – it's random
- (D) Sharing symmetric keys over email

Answer: B. Browsers/OSes come with pre-installed root CA certificates; if a CA in the chain is trusted, the server's identity is considered confirmed.

117. **Put the TLS handshake steps in the correct order:**

- (A) ServerHello -> ClientHello -> Key Exchange -> Certificate Validation -> Encrypted Data
- (B) ClientHello -> ServerHello+Certificate -> Certificate Validation -> Key Exchange -> Encrypted Communication
- (C) Key Exchange -> ClientHello -> Encrypted Communication -> ServerHello
- (D) Certificate Validation -> Encrypted Communication -> ClientHello -> ServerHello

Answer: B. Correct order: ClientHello -> ServerHello+Certificate -> Certificate Validation -> Key Exchange -> Encrypted Communication Starts.

118. **A Man-in-the-Middle (MITM) attack involves:**

- (A) The server attacking its own database
- (B) An attacker sitting between client and server, capturing/modifying traffic

- (C) A client sending too many requests
- (D) A router simply forwarding packets normally

Answer: B. In a MITM attack, an attacker positions themselves between client and server to intercept or alter the communication.

119. **HTTPS defends against MITM attacks mainly through:**

- (A) Faster transmission delay
- (B) Certificate validation – the browser checks that a trusted CA actually signed the certificate
- (C) Using more submarine cables
- (D) Statistical multiplexing

Answer: B. If an attacker presents a fake certificate, it won't have a valid trusted-CA signature, so the browser shows a 'Not Secure' / connection warning, blocking the MITM.

120. **In the HTTPS layering stack, where does TLS sit?**

- (A) Below the Physical layer
- (B) Between the Application layer (HTTP) and the Transport layer (TCP)
- (C) Above the Application layer only
- (D) It replaces the Network layer entirely

Answer: B. HTTPS stack (top to bottom): Application (HTTP) -> TLS (encrypts/authenticates) -> Transport (TCP) -> Network (IP) -> Link/Physical.

Part B: Scenario-Based Written Questions (20 Questions)

Each question below is the kind of applied, cross-lecture scenario the syllabus flags as high priority. Write your own answer first, then compare against the model solution given directly below each question.

1. **Netflix starts buffering heavily every night at 10 PM when millions of users stream simultaneously. Using concepts from ALL six lectures, explain (a) what is likely happening, and (b) at least three infrastructure-level fixes Netflix could use.**

(a) At peak time, demand from millions of concurrent users likely exceeds the capacity of one or more links or servers along the path – the end-to-end throughput is limited by the *bottleneck* link/server (Lecture 2: throughput = min of all link rates). If a server's queue/buffer fills up, packets get dropped (packet loss), and TCP must retransmit, which is perceived as buffering.

(b) Fixes: (1) **Load balancing** (Lecture 4) – add more servers and distribute the millions of requests across them instead of overloading one server. (2) **Edge Locations / CDN caching** (Lecture 1 & 4) – cache popular content physically closer to users, reducing propagation delay (d/s) and offloading the origin servers. (3) **Modern HTTP versions** (Lecture 5) – serving video over HTTP/2 or HTTP/3 (QUIC) reduces Head-of-Line blocking so one slow/lost segment doesn't stall the whole stream. All of this still rides over an encrypted HTTPS/TLS connection (Lecture 6) for security, and ultimately every packet still experiences the four delay components (Lecture 2) end-to-end.

2. **A satellite internet link advertises 500 Mbps bandwidth, yet video calls over it still feel laggy with a noticeable delay. Explain why, using the correct formulas, and state what would (and would not) fix this problem.**

This is a classic case of **high bandwidth $\neq$ low latency**. Bandwidth (500 Mbps) affects only *transmission delay* ($d_{trans} = L/R$) – how fast bits are pushed onto the link. The laggy feeling comes from *propagation delay* ($d_{prop} = d/s$), which depends only on the physical distance the signal must travel and the signal speed ($\approx 3 \times 10^8$ m/s), *not* on bandwidth. A geostationary satellite sits about 35,786 km up, so a round trip (Earth $\rightarrow$ satellite $\rightarrow$ Earth) covers roughly 71,500 km, producing about 240 ms of unavoidable propagation delay – regardless of how high the bandwidth is.

Fix: Increasing bandwidth (R) further would NOT help, since it only reduces transmission delay, which is already small for typical packet sizes. What WOULD help is reducing distance d – e.g., using low-earth-orbit satellites (much closer to Earth) or terrestrial/fiber links, or using Edge Locations so at least non-call traffic is served from nearby caches.

3. **Trace the full journey of data when a user taps the Instagram app icon and a reel starts playing, naming the OSI/TCP-IP layers, key devices, and protocols involved at each stage.**

1. **Physical/Data Link (L1/L2):** The phone converts the tap into a request; WiFi/4G radio signals (Physical layer bits) carry Ethernet/WiFi frames (Data Link, MAC addressing) to the local **router** (a L1–L3 wireless device).
2. **Network layer (L3):** The **router** uses IP addressing to forward the packet toward the ISP, and onward through the **network core** (many routers, each only knowing the next hop – store-and-forward, Lecture 2) to Instagram/Meta's data centers.
3. **Transport layer (L4):** TCP establishes a reliable connection (handling any packet loss/retransmission).
4. **Security (TLS):** A TLS handshake (Lecture 6) authenticates Meta's server via its certificate and negotiates a symmetric session key so the HTTP traffic is encrypted (HTTPS).
5. **Application layer (L7):** An HTTP(S) GET request (Lecture

5) is sent; on Meta's side, a **load balancer** routes it to an available server, possibly behind a **firewall** checking IP/port. The response (real data) is encapsulated back down the stack (adding TCP segment header, IP packet header, then a frame header) and travels back the same layered path, being decapsulated at each hop, arriving in well under a second thanks to Edge Locations/CDNs reducing propagation delay.

4. **HTTP is stateless, yet an online shopping cart 'remembers' items as you browse different pages. Explain how this works, and describe one conceptual similarity to how TLS maintains a 'session'.**

HTTP itself treats every request independently – the server does not automatically recall previous requests (Lecture 5). To make the cart persist, the server issues a **cookie/session token** the first time you interact with the site; the browser stores it and automatically attaches it to every subsequent request. The server then looks up that token's associated cart data server-side (or reads cart data directly from the cookie), giving the illusion of a continuous, stateful experience over a fundamentally stateless protocol.

Conceptual similarity to TLS: during the **TLS handshake** (Lecture 6), the client and server perform an expensive one-time exchange (asymmetric crypto) to agree on a shared *session key*. That key is then reused for all subsequent encrypted messages in the session, avoiding repeating the expensive handshake for every packet – just as a session token avoids re-authenticating on every HTTP request. Both are mechanisms for carrying 'state' across an otherwise per-message/per-connection process.

5. **A startup wants to connect 50 employee laptops directly to each other without any central device. (a) Calculate the number of cables required. (b) Explain why this is impractical, and (c) recommend a better solution with justification.**

(a) Using the direct-connection formula: $\text{Cables} = n(n-1)/2 = 50 \times 49/2 = \mathbf{1225}$ cables.
 (b) This is impractical because the number of required cables grows quadratically ($\sim n^2$) with the number of devices, while the number of devices grows only linearly. Managing, wiring, and troubleshooting 1225 physical connections for just 50 laptops is extremely costly, physically unwieldy, and does not scale – adding even a few more employees would sharply increase the cabling needed.
 (c) The recommended solution is to connect all 50 laptops to a central **switch** (Star topology). This needs only 50 cables (one per laptop to the switch) instead of 1225. The switch operates at Layer 2, learns MAC address-to-port mappings, and forwards frames only to the intended recipient, avoiding unnecessary broadcast traffic (unlike a hub). The trade-off is that the switch becomes a single point of failure, but for a small office this is an acceptable, well-justified choice (per the notes' environment-to-topology mapping: small office $\rightarrow$ Star).

6. **Explain the 'Dadi wali call' (phone call) analogy for circuit switching vs. packet switching, and describe a real-world scenario where circuit switching's guarantees would still be preferred over packet switching's efficiency.**

In the analogy: on an old landline (circuit switching), once Dadi's call connects, the entire path/capacity is reserved for the whole call – even if she pauses for 10 seconds, that capacity sits idle and cannot be used by anyone else (wasteful, but predictable/guaranteed). On a WhatsApp (packet-switched) call, if she pauses, no packets are sent during that pause, so that network capacity becomes instantly available for someone else's traffic – efficient, but with no guarantee that your next packet won't be delayed or dropped if the network is busy.
 Real-world scenario favoring circuit-switching-like guarantees: a hospital's dedicated emer-

gency communication line, or a live financial trading terminal that requires strictly predictable, jitter-free bandwidth. Even in modern networks, this is why techniques like reserved bandwidth/QoS (quality of service) circuits are sometimes still used for mission-critical, latency-sensitive, always-on traffic where the guarantee of stable capacity outweighs the inefficiency of occasional idle reservation.

7. A router receives 60% of a large packet's bits, but then the sending link fails and no more bits arrive. Per the store-and-forward rule, what does the router do with the partial data it has, and why is this rule important for how routers operate?

Per the store-and-forward rule, a router must receive the *entire* packet before forwarding even a single bit of it onward. If the link fails after only 60% of the bits have arrived, the router does **not** forward the partial data – since the packet was never fully received, it cannot be validated or completed, so the router discards/drops the incomplete packet. This will eventually be detected as packet loss by the transport layer (TCP) at the two end systems, which will trigger a retransmission of the entire packet.

Importance: store-and-forward means each router only ever needs to know how to reach the *next hop*, not the complete end-to-end path, since it fully buffers and checks each packet before passing it along. This simplifies router design and enables hop-by-hop, independent forwarding decisions – a core reason packet switching scales so well across the massively distributed 'network of networks' that is the Internet.

8. A 12,000-bit packet must be sent over a link with a rate of 3 Mbps, across a physical distance of 6×10^5 meters, with a signal speed of 2×10^8 m/s. Calculate the transmission delay, the propagation delay, and the total nodal delay (ignoring processing and queuing delay).

Transmission delay: $d_{trans} = L/R = 12000/3,000,000 = 0.004 \text{ s} = 4 \text{ ms}$

Propagation delay: $d_{prop} = d/s = 600,000/200,000,000 = 0.003 \text{ s} = 3 \text{ ms}$

Total (ignoring processing & queuing): $d_{trans} + d_{prop} = 4 \text{ ms} + 3 \text{ ms} = 7 \text{ ms}$

Note that transmission delay depends only on packet size and link rate, while propagation delay depends only on distance and signal speed – the two are independent physical quantities, both of which must be added to get the total delay experienced by this packet at this single hop (a full nodal delay would also add processing and queuing delay, which are out of scope for calculation in this course).

9. A data packet travels from your laptop to a cloud server through links of the following rates, in order: 1 Gbps (home WiFi) -> 100 Mbps (ISP access link) -> 40 Gbps (ISP backbone) -> 10 Mbps (a congested peering link) -> 25 Gbps (cloud data center network). What is the approximate end-to-end throughput, and what should be upgraded first to improve performance?

End-to-end throughput is approximately equal to the **bottleneck (minimum) link rate** along the path: $\min(1 \text{ Gbps}, 100 \text{ Mbps}, 40 \text{ Gbps}, 10 \text{ Mbps}, 25 \text{ Gbps}) = 10 \text{ Mbps}$ (the congested peering link).

Even though most of the path has very high capacity (up to 40 Gbps), the overall throughput a user experiences is capped by the single slowest link – 'the chain is only as strong as its weakest link.' To meaningfully improve performance, the **congested 10 Mbps peering link** should be upgraded or additional peering capacity/routes added first; upgrading any of the already-fast links (e.g., the 40 Gbps backbone) would have no effect on the end-to-end throughput, since it is not the bottleneck.

10. A packet travels from a laptop in Network A, through Network B, and finally to a server in Network C. Explain, layer by layer, the difference between what happens at Layer 2 versus what happens at Layer 3 during this journey.

Layer 3 (Network layer) represents the *whole journey*: the packet's source and destination IP addresses stay the same from the laptop in Network A all the way to the server in Network C. Routers at each network boundary use these IP addresses to decide the next network to forward the packet toward – this is the 'city to city' delivery described in the notes.

Layer 2 (Data Link layer) operates only *one hop at a time*: within Network A, the packet is wrapped in a frame addressed (via MAC addresses) to the next device on that specific LAN (e.g., the local router). Once it crosses into Network B, the Layer 2 frame is stripped and a brand-new frame with new MAC addresses is created for delivery within Network B, and again for Network C. So while the Layer 3 IP addressing is (largely) preserved end-to-end, the Layer 2 MAC addressing/frame is rebuilt at every hop – this is precisely why Layer 2 vs Layer 3 is flagged as the most commonly confused distinction in the course.

11. Using the gift-wrapping / nesting-doll analogy for encapsulation, explain what would go wrong (from a networking-design perspective) if a Layer 3 router tried to read and interpret the Layer 7 (Application) data inside every packet it forwarded.

In the gift-wrapping analogy, each layer adds its own wrapping (header) as data moves down the sender's stack, and the matching layer at the receiver removes only its own corresponding wrapping as data moves up – Layer 3 talks only to Layer 3, Layer 7 talks only to Layer 7 ('symmetry' between matching layers). Each layer is designed to understand only its own header, not the payload wrapped inside it.

If a Layer 3 router tried to read/interpret Layer 7 application data on every packet, several problems arise: (1) It would violate the entire purpose of layering – the router would need to understand every possible application protocol (HTTP, DNS, FTP, proprietary protocols, etc.), making router design and maintenance essentially impossible as new applications emerge. (2) It would massively slow down forwarding, since deep packet inspection of every byte is far more expensive than reading a small, fixed-format Layer 3 header. (3) It breaks the privacy/security model – if application data were unencrypted, a router in the path could read message contents (and if encrypted via TLS, it couldn't understand it anyway, defeating the point). This is exactly why layering exists: each device only needs to be 'intelligent' up to the layer it operates at (a router needs Layer 3, not Layer 7).

12. A small office network is experiencing frequent unnecessary traffic congestion because a legacy hub is being used to connect 20 computers. Diagnose the problem and recommend a specific device-level fix, explaining exactly why it solves the issue.

Diagnosis: A hub is a Layer 1 device with no addressing intelligence – it blindly broadcasts any incoming signal out to *all* other connected ports. With 20 computers, every single frame sent by any one computer is received by all 19 others, who must each inspect and then discard traffic not meant for them. This wastes the shared medium's capacity and causes congestion that worsens as more devices are added (this is explicitly the hub's core weakness in the notes).

Fix: Replace the hub with a **Layer 2 switch**. A switch learns which MAC address lives on which port (making it, per the notes, 'a hub with memory') and forwards each frame only out of the specific port where the destination device is connected, instead of broadcasting to everyone. This eliminates the unnecessary traffic that every other device previously had to receive and discard, directly resolving the congestion – while still requiring the same simple Star-topology cabling (each computer connects to one central device).

13. You are designing network topologies for three different scenarios: (1) a 10-person startup office, (2) a smart home with 15 IoT devices, (3) a large enterprise data center. Recommend a topology for each and justify your choice using fault tolerance and scalability trade-offs.

(1) 10-person startup office – Star topology. Easy to set up, easy to debug (every device connects to one central switch), and scalability is good for a small, predictable number of devices. The single-point-of-failure risk (the switch) is an acceptable trade-off at this small scale and low criticality.

(2) Smart home with 15 IoT devices – Mesh topology (often wireless mesh). IoT devices (bulbs, cameras, sensors) benefit from very high fault tolerance and multiple redundant direct paths – if one device or link fails, others can still communicate. The higher cabling/connection overhead ($n(n-1)/2$) is far less of a concern in a wireless mesh, and reliability of individual smart-device communication matters more than raw scalability here.

(3) Large enterprise data center – Hybrid topology. Data centers combine multiple topology types to get the benefits of each (e.g., high fault tolerance sections combined with manageable, scalable star-like clusters). Real-world large networks are hybrids precisely because different parts of the organization (storage, compute, edge-facing services) have different fault-tolerance and scalability requirements that no single pure topology satisfies well – despite being harder to manage, it is the standard choice at this scale.

14. An application is deployed across two Availability Zones (AZ-1 and AZ-2) within the same AWS Region. A flood knocks out AZ-1's power completely for six hours. Explain, using correct AWS terminology, why the application likely stays online, and what would have happened if it had only been deployed in a single AZ.

An Availability Zone is one or more discrete data centers within a Region, built with redundant power and networking and designed as an **isolated fault domain** – meaning a failure (like a flood) in one AZ is not supposed to directly affect other AZs in the same Region. Because the application is deployed across AZ-1 and AZ-2 (a **multi-AZ deployment**), when AZ-1 loses power, traffic and workloads can continue running on the healthy AZ-2, so the application as a whole stays available to users – this is exactly the fault-tolerance benefit multi-AZ deployment is meant to provide.

If the application had been deployed only in the single AZ that flooded, there would be no isolated, healthy AZ to fall back on – the loss of power in that one AZ would take the entire application offline for the full six-hour outage, since all its data centers/instances would be affected simultaneously. This illustrates why the notes emphasize 'Multi-AZ deployment = fault tolerant.'

15. Write out a complete, correctly formatted raw HTTP request that a browser might send to fetch a user's profile (GET /api/users/42) from api.example.com, and a plausible raw HTTP response for a successful JSON result. Label each part (request line, headers, body / status line, headers, body).

Request:

GET /api/users/42 HTTP/1.1	<- Request Line (Method, URL, Version)
Host: api.example.com	<- Header
User-Agent: Mozilla/5.0	<- Header
Accept: application/json	<- Header
	<- blank line (separates headers from body)
	<- (no body -- GET requests normally have none)

Response:

```

HTTP/1.1 200 OK                <- Status Line (Version, Code, Message)
Content-Type: application/json  <- Header
Content-Length: 47             <- Header
Date: Fri, 04 Sep 2026 00:00:00 GMT <- Header
                                <- blank line
{"id":42,"name":"Asha","role":"admin"} <- Body (actual JSON data)
Notes: GET is a safe, side-effect-free method (Lecture 5), so no request body is included. The
200 status code confirms success (2xx range); Content-Length tells the client how many bytes
of body to expect, and Content-Type tells it how to parse the body (as JSON here).

```

16. A web page has 4 resources to load: a large hero image, a CSS file, a JavaScript file, and a small icon. The hero image is unusually slow to load from the server. Compare what happens to the other 3 resources under HTTP/1.1, HTTP/2, and HTTP/3.

HTTP/1.1: Responses must arrive in strict FIFO order over the connection(s) in use. If the slow hero image is requested first (or is queued ahead of the others on a given connection), the CSS, JS, and icon responses behind it are forced to wait – this is classic **application-level Head-of-Line (HoL) blocking**. (Browsers partly work around this by opening several parallel TCP connections, but this adds overhead and doesn't fully solve the issue.)

HTTP/2: All four resources are requested as separate **streams multiplexed over a single TCP connection**, so at the application level, the CSS/JS/icon are no longer stuck waiting in a strict queue behind the image – they can be delivered as their data becomes ready. **However**, because it's still one underlying TCP connection, if the packet carrying part of the slow image causes TCP-level packet loss/retransmission, ALL streams (including CSS, JS, icon) can stall waiting for that TCP-level recovery – **TCP-level HoL blocking** still exists.

HTTP/3 (QUIC/UDP): Each stream is **truly independent** at the transport level. If the hero image's stream experiences loss and needs retransmission, only that stream is delayed – the CSS, JS, and icon streams continue delivering normally, unaffected. This is why HTTP/3 is described as having fully solved HoL blocking.

17. A frontend developer needs to let users change only their email address (not their entire profile) via a REST API. Which HTTP method should be used, and why would using PUT instead be a design mistake here? Give a brief example request.

The correct method is **PATCH**, since PATCH is designed for partial updates – modifying only specific fields of a resource (CRUD: Update, partial). Example:

```

PATCH /api/users/42 HTTP/1.1
Host: api.example.com
Content-Type: application/json

```

```

{"email": "new_email@example.com"}

```

Using **PUT** instead would be a design mistake because PUT semantically means *replace the entire existing resource* with what's provided in the request body. If the client sends only the email field via PUT, a strictly-compliant server implementation could interpret this as replacing the whole user record with just that one field – silently wiping out the user's other profile data (name, role, etc.) that wasn't included in the request. PATCH correctly signals 'update just this field, leave the rest alone.'

18. An attacker sets up a fake public WiFi hotspot named 'Free_Airport_WiFi' to intercept traffic from unsuspecting users (a Man-in-the-Middle setup). Explain step by step why a user visiting their HTTPS banking site is still protected, even though their traffic

physically passes through the attacker's device.

1. Even though the attacker's device physically routes/forwards the traffic, the user's browser still initiates a **TLS handshake** directly with the bank's actual server identity requirement – ClientHello is sent, expecting a ServerHello + Certificate in response.
2. The attacker cannot present the bank's real certificate because they don't possess the bank's **private key** (only the bank does) – the private key is never transmitted over the network. If the attacker forwards the bank's real certificate without being able to complete the corresponding private-key-based key exchange, the handshake will fail cryptographically.
3. If instead the attacker substitutes their own fake certificate for the bank's domain, the browser performs **certificate validation**: it checks whether a trusted **Certificate Authority** has signed that certificate for that exact domain. Since the attacker's self-generated certificate is not signed by any CA that the browser/OS already trusts (the pre-installed root CAs), validation fails.
4. The browser then displays a '**Not Secure**' / **connection warning**, alerting the user that the identity of the site could not be verified – stopping the user from unknowingly sending banking credentials to the attacker.
5. Even if the handshake somehow proceeded, all subsequent data is protected by **confidentiality** (symmetric session-key encryption) and **integrity** (so the attacker cannot silently read or tamper with the payload) – the combination of certificate-based authentication and encryption is what makes HTTPS resistant to this MITM setup, unlike plain HTTP where the attacker could read everything in plain text.

19. Walk through the complete TLS handshake, step by step, as it would occur when a user logs into their online banking website, explicitly stating which step prevents impersonation of the bank.

Step 1 – ClientHello: The user's browser sends the TLS versions it supports (e.g., TLS 1.3/1.2), a list of supported cipher suites, and a random nonce to the bank's server.

Step 2 – ServerHello + Certificate: The bank's server responds, selecting a cipher suite and sending its **digital certificate**, which contains the bank's public key and a signature from a Certificate Authority.

Step 3 – Certificate Validation (Authentication): The browser validates the certificate chain – checking that a CA it already trusts (via pre-installed root certificates) actually signed this certificate for this exact domain. **This is the step that prevents impersonation:** if the certificate is invalid, expired, self-signed, or doesn't match the domain, the browser shows a 'Not Secure' warning and the handshake effectively fails to establish trust, blocking any impersonator (e.g., a MITM attacker) from posing as the bank.

Step 4 – Key Exchange: Assuming validation succeeds, the client and server use asymmetric cryptography to securely derive a **shared symmetric session key**, without ever transmitting that key in a way an eavesdropper could capture it.

Step 5 – Encrypted Communication Starts: All further data – including the user's login credentials – is encrypted and decrypted using the fast symmetric session key, giving both confidentiality (encryption) and integrity for the rest of the banking session.

20. A basic packet-filtering firewall has the following rule: 'Allow inbound traffic only if Destination Port = 443 AND Destination IP = 10.0.0.5.' For each of the following incoming packets, state whether the firewall allows or blocks it, and why: (i) Dest IP 10.0.0.5, Dest Port 443. (ii) Dest IP 10.0.0.5, Dest Port 80. (iii) Dest IP 10.0.0.9, Dest Port 443. (iv) Dest IP 10.0.0.9, Dest Port 22.

A basic packet-filtering firewall (Lecture 4) inspects Layer 3 (source/destination IP) and Layer 4 (source/destination port) information – here, the rule requires BOTH conditions to be true.

(i) Dest IP 10.0.0.5, Dest Port 443: Allowed. Both conditions match – correct destination IP and the HTTPS port (443).

(ii) Dest IP 10.0.0.5, Dest Port 80: Blocked. The IP matches, but the port is 80 (plain HTTP), not the required 443 – the AND condition fails.

(iii) Dest IP 10.0.0.9, Dest Port 443: Blocked. The port matches, but the destination IP (10.0.0.9) is not the allowed 10.0.0.5 – the AND condition fails.

(iv) Dest IP 10.0.0.9, Dest Port 22: Blocked. Neither condition matches (wrong IP and wrong port, 22 being SSH, not HTTPS).

This illustrates that a basic firewall only reads two layers deep (IP + Port) – it does not inspect the actual application content, and both the AND'd conditions must hold true for traffic to be allowed through.

Final Quick-Recall Panel

- **Delays:** $d_{trans} = L/R$ $d_{prop} = d/s$ Throughput = min(all link rates) Cables = $n(n-1)/2$
- **Devices by layer:** Hub=L1 (broadcast) → Switch=L2 (MAC) → Router=L3 (IP) → Firewall=L3/L4 (IP+Port) → Load Balancer=L4/L7 (server choice)
- **PDU (L7-L1):** Message/Data → Data → Data → Segment → Packet → Frame → Bits
- **CRUD:** GET=Read, POST=Create, PUT=Replace, PATCH=Partial Update, DELETE=Delete
- **HTTP evolution:** 1.0 (new conn/file) → 1.1 (keep-alive, HoL blocking) → 2 (binary, multiplexed, TCP-level HoL remains) → 3 (QUIC/UDP, HoL fully solved)
- **TLS:** Asymmetric = secure handshake/key exchange; Symmetric = fast bulk data encryption; CA = signs/verifies certificates; Handshake order = ClientHello → ServerHello+Cert → Cert Validation → Key Exchange → Encrypted Communication
- **OSI mnemonic (top-down):** All People Seem To Need Data Processing

All the best for your test – CSAI321 Computer Networks.